Lab 4 — Pendulum: timing many swings

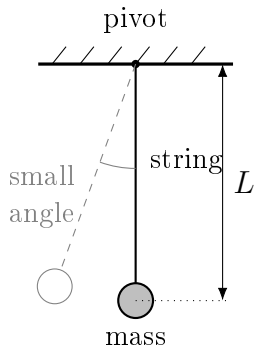
Class 1A — 28 October 2026

Name: _____ **Group:** _____ **Role now:** _____

Aim. Find out how the time for one swing, T, depends on the length L of the pendulum. Today we only *collect* the data. Write every number in the table on the back, even the strange ones.

What to do (*1 swing* = there and back)

1. **Measure** L from the pivot to the *centre* of the mass. Write it in the table.
2. **Pull** the mass sideways by a small angle (about 10°). Same angle every time.
3. **Release.** Start the stopwatch when the mass passes the lowest point. Count 20 swings aloud. Stop.
4. **Write** t_1 (time for 20 swings). Do it again and write t_2.
5. If t_1 and t_2 differ by more than 1 s, do a **third run** and write t_3.
6. **Change** L (use about 20, 30, 40, 50, 60, 80, 100 cm) and **swap roles**.



Roles: Release & count · Timer · Recorder · Measurer.

Variables. *Word bank: length of string L · time for 1 swing T · mass of the bob · starting angle*

Independent variable (I change it): _____

Dependent variable (I measure it): _____

Controlled variables (kept the same): _____ and _____

Warm-up (before we start). Time *one single swing* of a 50 cm pendulum: _____ s.

Prediction. How many swings do I need to time so that the timing error is under 1 %?

Before: _____ After the class: _____

Results — Pendulum

Group: _____

Smallest division of the ruler: _____ cm

Stopwatch reads to: _____ s

L (cm)	t_1 (s) 20 swings	t_2 (s) 20 swings	t_3 (s) only if needed	mean t (s) *at home*	$T = t/20$ (s) *at home*

Check. *Circle* any result in the table that looks odd. Do not erase it.

Finished early? Repeat $L \approx 50\,\text{cm}$ once more: $t =$ _____ s.

One thing I would improve in the method:

_____